

DYPAC-VLA: REPRESENTATION-TO-ACTION FIDELITY FOR LOW-BIT VISION-LANGUAGE-ACTION MODELS

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ABSTRACT

Low-bit vision-language-action (VLA) deployment differs from conventional model quantization: compression changes the policy, and the changed policy changes the observations it encounters. A precision mask that lowers action error on a fixed calibration buffer is therefore evaluated on observations the deployed policy may never see. We formulate deployment as *byte-constrained policy configuration under a mask-induced rollout distribution*: choose the mask that maximizes task success under an exact storage budget, with the expectation taken over the rollout distribution the mask itself induces. The formulation separates two roles that tensor-level quantization merges. Representation-level evidence proposes where precision is valuable, while only complete-policy action evidence decides whether an assembled configuration is admissible. We instantiate this proposal-admission separation in DYPAC-VLA. Representation-Integrity-aware Precision Allocation (RIPA) combines relational and distributional representation damage with action-impact weights to propose a guarded layout. Full-Context Precision Protection (FCP) projects that layout into a byte-feasible configuration and evaluates complete candidates through D-PAC, which defines temporal policy deviation through accumulated bias, replanning continuity, and control transitions; pointwise action error cannot distinguish these behaviors, so admission rests on sequence-level deviation. DYRANGE-A8 then removes the remaining closed-loop source of fidelity loss: the activation-range shift induced by policy-changed inputs, handled by input-conditioned A8 ranges while precision stays frozen, with no gradient fine-tuning or runtime precision routing. Across 50 RoboCasa365 tasks and 2,500 episodes per headline configuration, DYPAC-VLA attains the highest success among the evaluated PTQ methods on both backbones. GR00T N1.5 reaches **54.0%** while reducing audited Linear-component storage by **2.22 $\times$** , versus 30.4% for approximately size-matched QuantVLA and 55.1% for FP16. On $\pi_{0.5}$, it reaches **27.7%** at **2.70 $\times$** audited component compression, and the maximal-rate endpoint of the same rate family retains 26.8% at 3.56 $\times$. With the precision mask and quantized weights held fixed, input-conditioned A8 improves success by 10.6 percentage points over pooled static-A8 calibration. Artifacts are available at <https://anonymous.4open.science/r/DyPAC-VLA>.

1 INTRODUCTION

Vision-language-action (VLA) models map visual observations and language instructions to robot actions (Brohan et al., 2023; Octo Model Team et al., 2024; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025; NVIDIA et al., 2025). Their multimodal backbones and iterative action generators make model storage a practical deployment constraint, and post-training quantization (PTQ) reduces this footprint without retraining (Xiao et al., 2023; Lin et al., 2024). The target, however, is not a smaller network but a compact policy that still completes tasks, and this distinction breaks the assumption on which standard PTQ evaluation rests. Low-bit VLA deployment is not conventional model quantization: compression changes the policy, and the changed policy changes the observations it will encounter. A mask that lowers action accuracy under a fixed calibration buffer is therefore evaluated on observations the deployed policy may never see, and a mask that looks

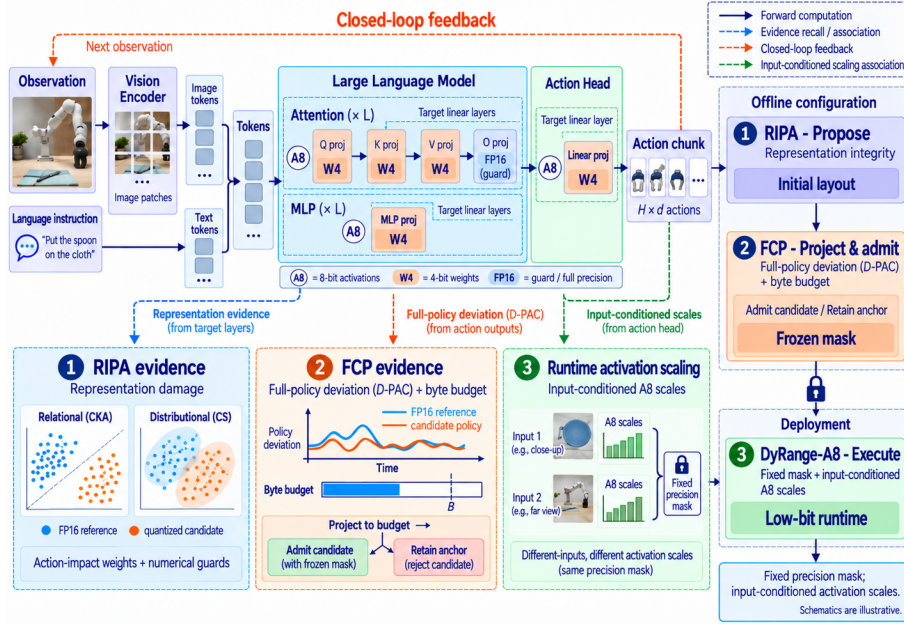


Figure 1: Three-stage representation-to-action quantization. RIPA compares relational and distributional damage and produces a guarded layout. FCP evaluates paired action sequences with D-PAC and component conditions, projects to the budget, and accepts an eligible candidate or retains the anchor. DyRange-A8 keeps the deployment mask fixed while adapting activation scales to each input. The diagram is schematic rather than a measured result.

admissible under fixed-buffer statistics can still alter accumulated bias, replanning discontinuities, and discrete control transitions once the policy acts.

We formulate this as *byte-constrained policy configuration under a mask-induced rollout distribution* (Equation 1): choose the layer-wise precision mask that maximizes expected task success under an exact storage budget, where the expectation is taken over the rollout distribution that the mask itself induces. Three consequences follow. The decision variable is a complete configuration, not an independent set of tensors, so simultaneous layer changes need not combine additively. The objective is task success on a mask-induced distribution, so layer-level proxies are proposals rather than decisions. The input distribution also moves with the policy, so activation handling belongs to the fidelity problem.

These consequences motivate a *proposal-admission separation principle*: representation-level evidence proposes where precision is valuable, but only complete-policy action evidence decides whether an assembled configuration is admissible. A favorable allocation score neither certifies a mixed-precision policy nor replaces evaluating the assembled model, and an admissible configuration may retain a byte-feasible anchor when no candidate qualifies. Admission also cannot rest on pointwise action error, because equal local error can induce different accumulated, replanning, and control-transition behavior.

We introduce DYPAC-VLA, a PTQ framework implementing these stages. First, *Representation-Integrity-aware Precision Allocation* (RIPA) combines relational and distributional representation damage, reweights it by action impact, and applies numerical guards and byte-constrained allocation. Second, *Full-Context Precision Protection* (FCP) evaluates interventions in the complete mixed-precision policy. Its D-PAC defines temporal policy deviation through accumulated bias, replanning continuity, and gripper events; complementary deviation conditions govern whether a complete candidate can replace a byte-feasible anchor. Third, DYRange-A8 executes the frozen layout with second-order four-bit weights (W4) and input-conditioned eight-bit activations (A8), removing closed-loop activation-range shift while keeping precision fixed. Dynamic behavior concerns activation scales, not precision routing or gradient fine-tuning.

Under the shared RoboCasa365 50-task protocol, DYPAC-VLA attains the highest observed success among the evaluated PTQ methods on both backbones. GR00T N1.5 reaches 54.0% success at $2.22\times$ audited Linear-component compression, 23.6 percentage points above approximately size-matched QuantVLA and within 1.1 points of the 55.1% FP16 reference. On $\pi_{0.5}$, it reaches 27.7% at $2.70\times$ compression, compared with 24.8% for QuantVLA and 26.2% for FP16; these configurations are not all storage-matched. A 4,000-episode LIBERO comparison evaluates cross-benchmark transfer, while controlled ablations test allocation, budget projection, and execution separately. In the strongest component-level control, which fixes the mask, the quantized weights, and the seeds, input-conditioned A8 raises success by 10.6 points over a pooled static-A8 calibration.

Our contributions are threefold.

- (1) **Problem formulation and separation principle.** We formulate low-bit VLA deployment as byte-constrained policy configuration under a mask-induced rollout distribution, and argue for a proposal–admission separation in which layer-level evidence proposes precision while only complete-policy action evidence decides admission.
- (2) **Proposal and admission machinery.** RIPA converts representation integrity and action impact into a guarded, byte-feasible layout; D-PAC defines temporal policy deviation, and FCP uses it with complementary conditions for budget projection and complete-candidate admission, retaining a feasible anchor when none qualifies.
- (3) **Strong low-bit VLA results.** We demonstrate leading observed success among the evaluated RoboCasa365 PTQ baselines on two backbones, with high-compression operating points and LIBERO transfer, and assess allocation, configuration decisions, and activation-range construction separately.

2 RELATED WORK

Efficient VLA deployment. Generalist VLA policies combine multimodal encoders with continuous-control heads (Brohan et al., 2023; Octo Model Team et al., 2024; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025; NVIDIA et al., 2025). Compact architectures, token reduction, temporal reuse, and architecture–quantizer co-design reduce computation as well as storage (Shukor et al., 2025; Zhang et al., 2025; Yang et al., 2025; Xu et al., 2025; Fang et al., 2025). We retain the pretrained architecture and study its success–storage tradeoff; Appendix A.13 reports the backend’s memory benefit and latency cost.

Action-aware VLA quantization. QuantVLA adapts distribution transformations to language and diffusion-action stacks (Zhang et al., 2026); ActQuant allocates tensor precision using action contribution and action-aware curvature (Akbari et al., 2026); Ω -QVLA combines rotations with denoising-step scaling (Wang et al., 2026b); and DA-PTQ uses trajectory motion drift for mixed-precision VLA quantization (Xu et al., 2026a). These methods share our premise that compression should be judged by its effect on control, and several already use action-side signals to allocate precision. The difference is where the decision is made. RIPA’s representation-level allocation only proposes a layout; FCP projects it to an exact byte budget and admits a complete candidate solely on full-policy temporal evidence; input-conditioned execution handles the remaining activation shift. Local adaptations of ActQuant and DA-PTQ are separated from their source protocols.

Calibration and dynamic activation ranges. Weight–activation PTQ uses smoothing, calibration, or rotations (Xiao et al., 2023; Shao et al., 2024; Ashkboos et al., 2024; Lin et al., 2024; Liu et al., 2025; Sun et al., 2025), and diffusion PTQ addresses timestep-dependent ranges and iterative error accumulation (Shang et al., 2023; Li et al., 2023; Wu et al., 2024; Chen et al., 2025; Zhao et al., 2024). Input-dependent scaling is established in ZeroQuant (Yao et al., 2022) and ViDiT-Q (Zhao et al., 2025), and our per-channel max-abs estimator is likewise standard. We ask a behavioral question rather than a quantizer question: what happens to a frozen range table once the quantized policy changes the inputs it must cover.

Precision allocation and behavioral proxies. Vision–language PTQ studies compensation, modality-aware channels, token saliency, and cross-layer dependencies (Deng & Huang, 2026; Xue et al., 2026; Wang et al., 2024; 2026a). Mixed-precision allocation uses hardware feedback,

Hessian sensitivity, reconstruction loss, or diffusion-specific proxies (Wang et al., 2019; Dong et al., 2019; 2020; Hubara et al., 2021; Li et al., 2021; Zhao et al., 2024). RIPA combines established relational and distributional statistics (Kornblith et al., 2019; Yin et al., 2026) with action-impact weights, numerical guards, and byte costs, without assuming that additive layer costs capture all interactions or that reference fidelity guarantees success.

The decision-level gap: tensor optimization versus complete-policy configuration. The literature above optimizes a fidelity proxy—reconstruction error, Hessian sensitivity, salience, or per-tensor bit width—and then deploys the resulting tensor-level solution. That pipeline inherits two assumptions that a closed-loop policy violates: that proxy improvements compose when layers change together, and that fidelity to a reference under a fixed buffer predicts behavior under the distribution the quantized policy induces. Work on input-conditioned computation shows that deployment-time gating and reuse change what the model sees (Xu et al., 2025; Zhang et al., 2025), and that acceleration gates can reduce closed-loop reliability without a clear action-level signature. We treat tensor-level evidence as a proposal mechanism and place the decision at the level of the complete configuration, evaluated on paired action sequences against the anchor the budget can support.

3 METHOD

Overview. DYPAC-VLA constructs a low-bit policy through three stages: RIPA proposes precision from representation integrity, FCP projects and admits configurations using action-sequence fidelity, and DYRange-A8 adapts activation ranges to the execution input. The output is one frozen precision mask and one input-conditioned activation rule. The stages follow a *proposal-admission separation*: layer-level evidence proposes precision, and only complete-policy action evidence decides admission. Appendix A.12 gives the full projection, adjudication, and freeze rules.

Problem formulation: byte-constrained policy configuration under a mask-induced rollout distribution. Let $f_\theta(o, \epsilon)$ denote a frozen FP16-reference VLA that maps observation o and generative noise ϵ to an action chunk $A \in \mathbb{R}^{H \times d}$, with target linear layers indexed by $\mathcal{L}$. A mask $M \in \{4, 16\}^{|\mathcal{L}|}$ assigns signed-nibble W4 or native 16-bit precision to each target, and $Q_M(f_\theta)$ denotes the mixed-precision policy under a fixed activation rule; FP16 follows the reference-row naming used throughout. Unlike static model compression, the deployed object is a policy whose actions change the states and observations that later queries are conditioned on. We therefore write $\mathcal{D}_M$ for the *mask-induced rollout distribution*, the rollouts that Q_M itself generates, and $S(Q_M; \tau) \in \{0, 1\}$ for task success on rollout τ . The deployment objective is

$$M^* = \arg \max_{M \in \{4, 16\}^{|\mathcal{L}|}} \mathbb{E}_{\tau \sim \mathcal{D}_M} [S(Q_M; \tau)] \quad \text{s.t.} \quad C(M) \leq C_{\max}, \quad (1)$$

where $C(M)$ is exact packed static-component storage. Equation 1 is the evaluation goal, not FCP’s selection rule: evaluating every mask’s rollout distribution is intractable. It also fixes the roles of the two evidence types. Because the decision variable is one coupled mask, simultaneous layer changes need not combine additively, so layer-level scores cannot serve as decisions; and because the distribution is mask-induced, fidelity on a fixed calibration buffer is a proposal signal rather than an admission certificate. FCP therefore screens complete policies on paired teacher contexts, checks shortlisted candidates on candidate-induced observations, and otherwise retains a feasible anchor (Section 3.2).

FCP is label-free: success and reward never enter its guards. Candidate and reference share generative noise, and an independent noise stream is reserved for post-freeze audit. The RIPA ratio is selected by a disclosed four-task success sweep, and deployment uses a global fixed mask without gradient updates, task-conditioned routing, output correction, or FP16 bypass.

3.1 RIPA: REPRESENTATION-INTEGRITY-AWARE PRECISION ALLOCATION

RIPA combines two damage views: centered linear kernel alignment (CKA) (Kornblith et al., 2019) measures input-corresponding relations, and the uncentered Gaussian-kernel Cauchy–Schwarz divergence used in CS-Aligner (Yin et al., 2026) measures distributional change, with bandwidth fixed from reference samples. The relational signal is recorded at the action generator’s output

representation (CKA-DiT); CS is measured at the intervened layer. The two views are complementary: a uniform translation of features preserves centered relations while shifting the distribution, and a row permutation preserves the distribution while disrupting input-corresponding relations (Appendix A.1). RMS and saturation guards protect numerically unstable layers separately.

Let $\bar{d}_i^{\text{rel}} = \text{norm}(1 - \widehat{\text{CKA}}_i)$ and $\bar{d}_i^{\text{dist}} = \text{norm}(\widehat{D}_{\text{CS},i})$ be min-max-normalized damage scores, with the normalizing constants taken jointly over all measured layer-bit pairs of the frozen score bank. RIPA scores each target and allocates precision under a byte budget:

$$\begin{aligned} \mathcal{I}_i &= w_i [\alpha \bar{d}_i^{\text{rel}} + (1 - \alpha) \bar{d}_i^{\text{dist}}], \\ M_{\text{init}} &\approx \arg \min_M \sum_i \mathbf{1}[M_i = 4] \mathcal{I}_i \quad \text{s.t.} \quad C(M) \leq C_{\text{init}}, \quad M_i = 16 \quad (i \in \mathcal{G}). \end{aligned} \quad (2)$$

Here w_i is an action-impact weight derived from one-layer D_{func} values, clipped to $[0.5, 2]$ before renormalization; $\mathcal{G}$ hard-protects layers exceeding the RMS or saturation thresholds; and C_{init} is the initializer’s budget, which can differ from FCP’s deployment ceiling C_{max} . A four-task development sweep over seven choices fixes $\alpha = 16/17$, a 16:1 relational-to-distributional ratio. Appendix A.1 gives the weight formula, deployed range, guard thresholds, and calibration provenance.

The $\pi_{0.5}$ port transfers the frozen 16:1 ratio and reruns target-architecture sensitivity and complete-mask adjudication under its earlier Uniform-W6 budget, producing an 80-W4/100-FP16 mask for FCP without copying GR00T layer identities. FCP then adapts that artifact to the tighter QuantVLA-relative ceiling; direct target-budget RIPA is not evaluated here.

3.2 FCP: FULL-POLICY BUDGET PROJECTION AND FIDELITY ADMISSION

Full-Context Precision Protection (FCP) turns representation-based proposals into discrete configuration decisions. It first retains or projects M_{init} to a byte-feasible anchor M_0 , then admits a complete candidate only if it meets the prescribed action-deviation conditions. D-PAC defines how to measure sequence deviation; FCP defines how that evidence governs precision changes.

D-PAC: defining temporal policy deviation. D-PAC is a definition of *temporal policy deviation* for configuration evaluation and admission, not a new scalar error metric. Pointwise action error cannot serve the admission decision, because equal local error can produce different accumulated, replanning, and control-transition behavior. Consider $e_t^{(1)} = \epsilon$ and $e_t^{(2)} = (-1)^t \epsilon$: both have per-step mean squared error (MSE) ϵ^2 , yet their sums over even T are $T\epsilon$ and zero. Equal local errors can therefore have different temporal structure, so the decision needs a sequence-level definition.

D-PAC scores paired physical sequences. For each task-seed sequence, teacher and candidate share observations and initial action noise; we inverse-normalize the final chunks and retain all $H = 16$ actions executed before each replan, without overlapping forecasts. Errors use a robust coordinate scale estimated globally from FP16 data, so a candidate cannot alter its denominator. Five normalized components address three control questions and receive equal weight:

$$\begin{aligned} d_{\text{PAC}}^{(u)} &= \frac{1}{5} (d_{\text{local}} + d_{\text{prefix}} + d_{\text{pose}} + d_{\text{stitch}} + d_{\text{grip}}), \\ D_{\text{PAC}} &= \text{mean}_u d_{\text{PAC}}^{(u)} + \text{CVaR}_{0.9,u} (d_{\text{PAC}}^{(u)}). \end{aligned} \quad (3)$$

The three components d_{local} , d_{prefix} , and d_{pose} measure persistent deviation through local error, prefix accumulation, and right-composed SE(3) pose deviation; d_{stitch} measures replanning continuity at chunk boundaries; and d_{grip} measures discrete control consistency. Here u indexes task-seed sequences, and conditional value at risk (CVaR) averages the largest $\lceil 0.1|u| \rceil$ sequence deviations, giving this tail the same total weight as the mean. We retain D_{func} for complementary final-action and kinematic evidence. FCP uses 12 stratified tasks, three seeds each and four replans: 36 sequences and 144 teacher observations, distinct from RIPA’s historical score bank. Appendix A.2 gives the formulas and edge cases.

Projection followed by admission. FCP projects the initial layout, generates exact-byte proposals, and admits complete candidates using task-level deviation bounds and candidate-induced observations. A feasible initializer becomes the anchor; nothing forces a precision change. Projection instantiates

each FP16-to-W4 removal as a complete policy and removes protections from least to most damaging until the mask fits, using the measured intervention $\Delta_i(M) = D(Q_{M^{(i)}}; \mathcal{B}) - D(Q_M; \mathcal{B})$ in the current mixed-precision background rather than as isolated layer reconstruction. On $\pi_{0.5}$ this removes 41 of 100 protections, producing the 121-W4/59-FP16 anchor.

Admission then flips target precision states one at a time from M_0 . Each intervention is instantiated as a mixed-precision network and compared with its complete-mask background, and the resulting conservative benefits feed an exact sparse Pareto 0–1 dynamic program subject to $C(M) = C_{\text{fixed}} + \sum_{i \in \mathcal{L}} C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}$. The constraint is expressed in bytes rather than layer count because layers differ in tensor and metadata size. The dynamic program only generates candidates: its additive benefits cannot capture interactions among simultaneous layer changes, so FCP evaluates every candidate as a complete configuration. A candidate is admitted only if the task-level minimax bound is strictly negative in every metric–group pair with non-positive component bounds for pose, stitch, and gripper; the rule preserves the anchor unless deviation evidence favors replacement, and extra compression alone cannot justify admission. On GR00T all proposals fail the teacher-state screen, so FCP retains M_0 and claims no success optimality. Appendix A.12 gives the minimax statistic, the candidate-state check, and the deterministic freeze order.

3.3 DYRANGE-A8: RESOLVING CLOSED-LOOP ACTIVATION-RANGE SHIFT

The deployment stage combines Hessian-aware W4 weights with standard per-forward max-abs A8 ranges, adapting scales to realized inputs while keeping precision fixed. Its design target is one failure mode rather than dynamic quantization in general: a closed-loop activation shift, in which the deployed policy changes the states and flow-step inputs that the ranges must cover.

Weight quantization. For each W4 layer, 256 calibration observations form FP16 input second-order statistics $X^\top X$. A GPTQ-style sequential error-feedback pass selects clipping and quantizes signed weights in groups of 64 (Frantar et al., 2023). Two signed four-bit codes are packed per byte; per-group scales and packing metadata are retained, but the FP16 weight tensor is not. Row rotation is the identity: the reported numbers require no learned transform.

Closed-loop activation-range shift. A static activation table is estimated once, under the observations, flow steps, and policy-induced states of the calibration buffer. Once the quantized policy acts, its inputs drift away from that buffer, and a frozen table becomes mismatched even though its calibration was correct at collection time. Our static counterfactual makes the shift concrete: independently calibrated 99.9th-percentile scales per target layer, input channel, and applicable flow step, evaluated under an otherwise identical runtime. DYRANGE-A8 removes the lookup and recomputes a scale for every target linear layer, forward call, and input channel:

$$s_c = \max\left(\frac{\max |x_{\dots, c}|}{127}, 10^{-6}\right), \quad q_c = \text{clip}(\text{round}(x_c/s_c), -127, 127). \quad (4)$$

The rule is deterministic and uses the input dtype, with no calibration lookup, flow-step selector, task selector, or candidate branch. It changes activation ranges only: precision stays frozen and no runtime routing occurs (Appendix A.13).

Exact storage scope. Storage counts the declared Linear-component inventory: fixed tensors within that inventory, packed W4, scales, and metadata. It excludes vision modules outside the inventory, activations, and runtime caches. GR00T’s 100-W4/16-FP16 layout occupies 962,068,480 component bytes; $\pi_{0.5}$ ’s 121-W4/59-FP16 layout occupies 1,634,828,288. Appendix A.17 distinguishes these counts from whole-model packs and alternate inventories; Appendix A.19 reports available offline work counts.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

We evaluate GR00T N1.5 and $\pi_{0.5}$ (NVIDIA et al., 2025; Physical Intelligence et al., 2025) on RoboCasa365 v1.0.1 (Nasiriany et al., 2026): 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks. Headline configurations use 50 seeds per task, the official horizon, four flow steps,

Table 1: RoboCasa365 success and audited Linear-component storage on both backbones (50 tasks; 2,500 episodes per headline configuration).

Configuration	Allocation	Atomic SR $\uparrow$	C-Seen SR $\uparrow$	C-Unseen SR $\uparrow$	All SR $\uparrow$	Linear size (GiB) $\downarrow$	Comp. $\uparrow$
$\pi_{0.5}$ Physical Intelligence et al. (2025) (<i>formal target split</i>)							
FP16	—	58.3	14.0	2.1	26.2	4.113	1.00 $\times$
QUANTVLA W4A8	180 W4	56.1	12.8	1.8	24.8	1.388	2.96 $\times$
Uniform W6	180 W6	56.6	10.9	2.1	24.5	1.902	2.16 $\times$
Ω -QVLA W4A4 [‡]	252 W4	49.6	10.4	1.0	21.5	1.307 ^L	—
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2-Q4	56.6	12.3	2.8	25.2	2.904 ^P	—
DA-PTQ W4A8	87 W4/15 BF16	58.6	12.3	3.8	26.2	1.390	2.96 $\times$
DYPAC-VLA (Ours)[†]	121 W4	59.6	15.4	4.3	27.7	1.523	2.70 $\times$
DYPAC-VLA (Ours, all-W4)[#]	180 W4	58.7	14.5	3.3	26.8	1.157	3.56 $\times$
GR00T N1.5 (<i>high-success control</i>)							
FP16	—	75.6	41.9	45.3	55.1	1.993	1.00 $\times$
QUANTVLA W4A8	116 W4	49.6	19.6	19.8	30.4	0.898	2.22 $\times$
Uniform W6	116 W6	68.4	42.9	41.8	51.7	1.109	1.80 $\times$
Ω -QVLA W4A4 [‡]	180 W4	60.1	25.9	27.5	38.7	0.599	3.33 $\times$
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2-Q4	72.1	42.6	41.5	52.9	3.655 ^P	—
DA-PTQ W4A8	55 W4/9 BF16	8.1	0.0	0.4	3.0	1.110	1.80 $\times$
DYPAC-VLA (Ours)[†]	100 W4	74.7	43.9	40.9	54.0	0.896	2.22 $\times$
DYPAC-VLA (Ours, all-W4)[#]	116 W4	70.9	42.9	42.3	52.8	0.779	2.56 $\times$

Linear size and compression count the audited Linear-component inventory of the same checkpoint and exclude runtime memory and latency. *P* marks a whole-model pack (GR00T: three-checkpoint mean) and *L* the distinct 252-Linear inventory; their common-scope compression is suppressed (Appendix A.17). C-Seen/C-Unseen denote Composite-Seen/Unseen. [†]Ours includes ratio-selection tasks. [#]Ours, all-W4 is the maximal-rate endpoint of the same DYPAC-VLA rate family, derived by the same FCP rule from our own pipeline, not an independent baseline. [‡] Ω -QVLA uses task-set-specific calibration. [§]ActQuant (Akbari et al., 2026) is a local adaptation (source-protocol-equivalent=false). ^{||}DA-PTQ (Xu et al., 2026a) is a local adaptation (source-protocol-equivalent=false).

and paired task–seed–replan noise (2,500 episodes each). Focused controls use seeds 0–9 unless specified. LIBERO evaluates framework applicability with separately calibrated suite-specific plans, fixed within each deployment.

Comparison and statistics. We compare FP16, QuantVLA (Zhang et al., 2026), Uniform W6, Ω -QVLA (Wang et al., 2026b), and local ActQuant/DA-PTQ adaptations. Storage counts the declared Linear-component inventory, including packing metadata and fixed tensors; the same substitution gives 1.18 $\times$ over the complete GR00T checkpoint payload versus 2.22 $\times$ within that inventory, and whole-model packs and alternative inventories are marked separately (Appendix A.17). We report task-macro success and paired two-sided McNemar tests with Holm correction within registered families; non-significance establishes neither equivalence nor non-inferiority. RIPA’s four ratio-selection tasks remain in the headline; on the other 46 tasks, success is 51.8/27.6/52.8% for ours/QuantVLA/FP16. Frozen manifests bind masks, code, and coverage.

4.2 ROBOCASA365: SUCCESS-STORAGE RESULTS

DYPAC-VLA attains the highest observed success among the evaluated PTQ methods on both backbones. GR00T reaches 54.0% at 2.22 $\times$ audited Linear-component compression, 23.6 points above the approximately size-matched QuantVLA (Holm $p < 10^{-4}$) and 1.1 points below FP16 (55.1%, $p = 0.2929$). On $\pi_{0.5}$, it reaches 27.7% at 2.70 $\times$, versus 24.8% for QuantVLA and 26.2% for FP16; that margin is descriptive and the rows are not all size-matched. The all-W4 rows are the maximal-rate endpoint of the same DYPAC-VLA rate family, not a separate baseline: the same FCP rule produces them from the same inventory, and comparing them with the headline mask on success would test two operating points of one procedure against each other. Their role is to show the range the pipeline covers. One rule yields 54.0% at 2.22 $\times$ and 52.8% at 2.56 $\times$ on GR00T, and 27.7% at 2.70 $\times$ and 26.8% at 3.56 $\times$ on $\pi_{0.5}$; the appendix sweep extends the family to 1.17 $\times$ at 55.4%. What the allocation and admission stages contribute is therefore a budget-to-configuration mapping,

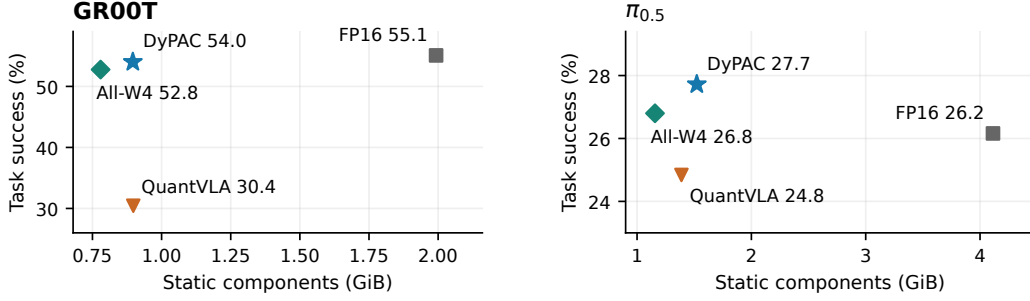


Figure 2: Selected RoboCasa365 success-storage operating points under each backbone’s common Linear-component accounting scope.

including the projection that makes an infeasible $\pi_{0.5}$ initializer feasible, not an established success gain over a simpler endpoint of the same family (Appendix A.5).

Adapted methods. ActQuant reaches 52.9/25.2% on GR00T/ $\pi_{0.5}$ (ours-versus-ActQuant Holm $p = 0.3102/0.0011$); DA-PTQ reaches 3.0/26.2% ($p = 0.0648$ on $\pi_{0.5}$). Both are local, source-protocol-nonequivalent architecture adaptations, so their outcomes characterize the adapters rather than the native source methods.

4.3 FRAMEWORK APPLICABILITY ON LIBERO

Table 2: LIBERO cross-benchmark transfer: four-suite mean success and audited Linear-component size on both backbones (10 tasks per suite; 100 episodes per suite).

Configuration	$\pi_{0.5}$		GR00T N1.5	
	Avg. SR $\uparrow$	Size (GiB) $\downarrow$	Avg. SR $\uparrow$	Size (GiB) $\downarrow$
FP16	96.8	4.113	86.8	1.993
QUANTVLA W4A8	96.3	1.388	81.3	0.898
Uniform W6	96.8	1.902	70.5	1.109
Ω -QVLA W4A4 [†]	98.0	1.307 ^L	84.5	0.599
DyPAC-VLA (Ours)	97.3	1.788	93.8	0.948

The joint local gate covers 4,000 episodes across 40/40 exact-valid suite cells. ^LThe $\pi_{0.5}$ Ω -QVLA size is a distinct 252-Linear inventory, not the common component scope (Appendix A.17). [†]The displayed $\pi_{0.5}$ Ω -QVLA success value is experiment-owner attested. Ours size is the arithmetic mean of four suite-specific static plans. Appendix A.20 reports the full suite breakdown and protocol.

The 4,000-episode matrix yields 97.3% on $\pi_{0.5}$ at 1.788 GiB and 93.8% on GR00T at 0.948 GiB. Ours leads the displayed GR00T rows; on $\pi_{0.5}$, Ω -QVLA reaches 98.0% at 1.307 GiB under its distinct inventory. The framework transfers; the mask does not, since each suite has a separately prepared plan (Appendix A.20).

4.4 RIPA AND FCP: ALLOCATION AND DECISION EVIDENCE

Independent signal comparison. An equal-byte integer program rebuilds all three RIPA variants from the same frozen bank, with a fused 93-W4/23-FP16 mask that differs from the headline 100-W4/16-FP16 mask. On Primary-46, fusion reaches 53.26%, versus 51.52% and 50.65% for relational-only and distributional-only allocation (both Holm $p = 0.5796$); the fused advantage is directional and unresolved (Appendix A.6).

Budget projection and admission. FCP removes 41 of $\pi_{0.5}$ ’s 100 initial protections, producing the feasible 121-W4/59-FP16 anchor from a mask exceeding the ceiling by 205,587,047 bytes. A 500-key diagnostic gives 28.2% versus 26.0% ($p = 0.235$), and an independent 7,000-episode comparison finds no resolved advantage over final-action-error or largest-first deletion (eight Holm $p = 1.0$). D-PAC passes three controlled matched-MSE sensitivity tests, yet MSE, D-PAC, and D-PAC plus

components reject all six GR00T candidates under strict one-SE improvement: on this candidate set, the sequence-level rule adds no admission benefit. The rejected proposals differ from the retained initializer by no Holm-significant margin (Appendices A.7, A.14, A.18, A.15).

4.5 CONTROLLED COMPONENT COMPARISONS

Table 3: GR00T activation-range comparison at a fixed 100-W4/16-FP16 mask (50 tasks; seeds 23000–23009; 500 episodes per arm).

Range construction	Atomic	C-Seen	C-Unseen	All (%)	Successes
Static: registered table	0.0	0.0	0.0	0.0	0/500
Static: two-row FP16 calibration	0.0	0.0	0.0	0.0	0/500
Static: pooled FP16 calibration	56.1	33.1	38.8	43.2	216/500
Dynamic: DYRANGE-A8	68.9	43.8	46.9	53.8	269/500

All rows share the 100-W4/16-native-precision layout, per-split Hessian W4 weights, 50 tasks, and fresh paired seeds 23000–23009. FP16 denotes the calibration source, not deployment weight precision. Dense and two-row tables use the same 256 calibration observations/noise and 99.9th-percentile rule.

At a fixed mask, fixed Hessian W4 weights, and paired seeds, only the activation-range rule changes: pooled static A8 reaches 43.2%, the matched two-row and registered tables reach 0%, and input-conditioned DYRANGE-A8 reaches 53.8%. DYRANGE-A8 exceeds the stronger pooled-static control by 10.6 points (Holm $p = 1.27 \times 10^{-4}$; 95% CI [6.2, 15.0]; Table 3), so sparse calibration sampling explains a substantial static-calibration failure while input-conditioned ranges retain a measured advantage over a working static baseline. The contrast does not isolate the policy-induced component of the shift. At the same fixed 100/16 mask, replacing DuQuant/static-A8 with Hessian-aware W4 and DYRANGE-A8 raises success from 50.8% to 54.0% (Holm $p = 0.0036$), while local-MSE and no-final-check selection each reach 53.0% versus 53.8% for the anchor under different masks and byte counts. Neither historical control establishes an independent benefit for RIPA’s allocation or FCP’s admission stages (Appendices A.10, A.8).

5 CONCLUSION AND LIMITATIONS

Low-bit VLA deployment is a policy-configuration problem: the mask changes the policy, and the policy changes the observations it meets. We formulated this as byte-constrained policy configuration under a mask-induced rollout distribution and instantiated a proposal–admission separation in DYPAC-VLA: RIPA proposes precision from representation integrity, FCP projects the layout to an exact byte budget and admits complete candidates only on full-policy temporal evidence, and DYRANGE-A8 resolves the remaining closed-loop activation-range shift with a frozen mask. On the shared RoboCasa365 evaluation, the method attains the highest observed success among the evaluated PTQ methods on both backbones: 54.0% at $2.22\times$ audited Linear-component compression on GR00T and 27.7% at $2.70\times$ on $\pi_{0.5}$. With the mask and quantized weights held fixed, input-conditioned A8 improves success by 10.6 points over a pooled static-A8 calibration.

Limitations. Fusion and projection-rule superiority remain statistically unresolved. Controlled probes verify the temporal sensitivities that D-PAC targets, while the matched-candidate replay selects the same anchor under all three criteria. The calibration inputs and frozen-bank allocation are reproduced; the historical candidate-ranking stage has not been regenerated, and complete offline configuration costs are unavailable. Evaluation is simulation-only, some baselines require non-equivalent adapters or different measurement scopes, and the generic backend trades memory for latency and energy. The demonstrated benefit is component-storage compression at competitive success; the whole-checkpoint ratio is smaller (Appendix A.17).

AI USE STATEMENT

Generative AI tools assisted with planning, implementation, debugging, aggregation, writing, formatting, and figure preparation. The authors reviewed all generated material, tested AI-assisted code, reran the evidence and document gates, and verified claims against frozen manifests and primary

sources. No experimental result was synthesized by a generative model; the authors take responsibility for the paper and its artifacts.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the method, evaluation, and statistical protocols. The supplement documents content-addressed plans, manifests, coverage checks, table generation, and claim–evidence audits. Headline cells are regenerated from frozen summaries.

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A SUPPLEMENTARY EVIDENCE

A.1 RIPA ALLOCATION DETAILS AND STATISTIC PROVENANCE

We construct the GR00T initializer with RIPA before FCP. In the historical implementation notation, $S_i = (\lambda_g \bar{d}_i^{\text{geom}} + \lambda_d \bar{d}_i^{\text{dist}}) / (\lambda_g + \lambda_d)$, where $\bar{d}_i^{\text{geom}} = \bar{d}_i^{\text{rel}}$ and $\bar{d}_i^{\text{dist}}$ are jointly normalized versions of $1 - \widehat{\text{CKA}}_i$ and $\widehat{D}_{\text{CS},i}$. Equation 2 writes the same cost as $\mathcal{I}_i = w_i S_i$. Action-impact weights prioritize layers whose one-layer intervention changes physical actions, while RMS and saturation guards retain unstable layers in FP16. A greedy byte-constrained solver followed by fixed-size flips produces nine complete candidate masks. Paired adjudication on 16 calibration observations selects one mask for each weighting without success labels.

Normalization and action weights. For each statistic $z \in \{1 - \widehat{\text{CKA}}, \widehat{D}_{\text{CS}}\}$, let $\mathcal{P}$ contain all finite measured (i, b) pairs in the frozen score bank. The selector computes $\bar{z}_{i,b} = (z_{i,b} - \min_{\mathcal{P}} z) / (\max_{\mathcal{P}} z - \min_{\mathcal{P}} z)$ independently for the two terms, setting a constant term to zero when its range is below 10^{-12} . Missing terms are omitted and the available fusion weights renormalized; a target without a usable score remains FP16. The deployed score uses the W4 entries. The relational field is `cka_dit` (action-generator output), whereas CS is measured at the intervened Linear output.

The recorded action-impact source is the one-layer functional deviation $d_i = D_{\text{func},i}$, not raw MSE or D-PAC. With n targets, missing d_i is replaced by the mean of measured values; if none are measured or their sum is nonpositive, $w_i = 1$. Otherwise,

$$v_i = \frac{nd_i}{\sum_j d_j}, \quad c_i = \text{clip}(v_i, 0.5, 2), \quad w_i = \frac{nc_i}{\sum_j c_j}. \quad (5)$$

The final renormalization preserves mean one but rescales the clipping endpoints. The frozen GR00T bank gives $w_i \in [0.76535, 3.06141]$, not $[0.5, 2]$. Its raw D_{func} values range from 5.17×10^{-5} to 0.32739.

Why the two damage views are complementary. For paired feature matrices, $Y = X + 1b^\top$ preserves centered relations while translating the distribution. Conversely, $Y = PX$ for a row permutation P preserves the empirical distribution but generally disrupts input-corresponding relations. Each view is therefore blind to a transformation the other detects, which is why RIPA scores both.

Hard numerical protection. For reference/quantized pooled outputs X, Y , the RMS guard is $r_i = \text{median}_c |\log((\text{RMS}(Y_c) + 10^{-6}) / (\text{RMS}(X_c) + 10^{-6}))|$. The saturation guard is the fraction of elements of Y exceeding $P_{99.9}(|X|)$ in magnitude. Thresholds are 1.5 times the nearest-rank 99th percentile of measured W4 guard values, unless supplied by the frozen bank. GR00T uses $\tau_{\text{rms}} = 0.1318639107$ and $\tau_{\text{sat}} = 0.0023243131$; exceeding either removes a layer’s quantization options and retains it in FP16. The protected numerical outlier is language layer 2’s MLP down-projection. These are feasibility cuts, not soft penalties.

Historical calibration provenance. The headline initializer’s parent plan and the independent signal comparison read `checkpoints/packs/robocasa365/sensitivity_robocasa365_atomic_protocolfix_d4_g64_b4_6_dit.json` (SHA-256 prefix `29e1a5c7a991e2ab`). Its calibration buffer is deterministic and self-contained: regenerating it from seed 0, 256 observations, horizon 16, action dimension 32, and the RoboCasa365 observation format reproduces the recorded digest `1f61d740...f531c` exactly. The digest covers the generation parameters and the observation and noise tensors, so it identifies the inputs rather than only the configuration. Those inputs are synthetic and data-free—random images and states drawn from the checkpoint’s recorded ranges—in RoboCasa365 format. The bank’s `obs_source` constant instead names the L1 format; both are synthetic, so only that format label disagrees with the buffer the digest fixes. The same digest appears in the static A8 range table and in the allocation report, linking the bank, that table, and the archived final mask to one buffer. The 256-observation buffer is the A8 calibration set; the representation statistics use its first 16 observations, and the action-impact weights use the first eight with two paired noise draws each. We neither relabel the bank as FCP’s 12-task on-policy buffer nor present FCP’s buffer as its source.

Protocol attribution. The historical 54.0% configuration was produced under the v2.0 protocol: the frozen 100-W4/16-FP16 mask, its Hessian W4 construction, and the DYRANGE-A8 execution stack are those of the v2.0 pipeline. The bank records a wrapped representation reference with FP16 target-layer weights, static A8 activation scaling, and rotations active, while its functional pairing and guard reference use the pure FP16 model; we report those archived fields as recorded.

What the audit establishes. The audit reproduces the recorded calibration inputs and regenerates all nine precision candidates from the frozen sensitivity bank. Re-running the archived allocation entry point from that bank, with the inputs recorded in the parent plan, reproduces the parent plan’s 116-layer precision vector, its byte total, and all nine candidates exactly; the archived deployment mask matches candidate 2 (historical label `flip8`). The audit does not recompute the sensitivity bank and does not rerun the historical candidate-ranking stage. The shared A8 range table belongs to the historical static-activation path; the deployed DYRANGE-A8 rule recomputes ranges per forward and does not read it. Table 4 records each link; the audit is reproduced by `scripts/tools/verify_ripa_provenance.py` and stored at `docs/gdsq_vla_iclr2027/evidence/ripa_provenance_audit.json`.

Table 4: Historical RIPA score-bank provenance chain.

Link	Surviving record	Status
Calibration inputs	Deterministic synthetic buffer, seed 0, 256 observations, horizon 16, action dimension 32, RoboCasa365 format; the digest covers generation parameters and input tensors	Re-derived exactly
Buffer reuse	The same digest appears in the static A8 range table and the allocation report, tying the bank, that table, and the final mask to one buffer	Confirmed
Metadata field	<code>obs_source</code> names the L1 format; the buffer is RoboCasa365-format synthetic data, so only the format label differs	Format label only
Reference path	v2.0 protocol; the bank records a wrapped representation reference with FP16 target-layer weights, static A8 activation scaling, rotations active, and pure FP16 for the functional pairing and guards	v2.0
Measurement location	Relational CKA at the action-generator output (<code>cka_dit</code>); distributional CS at the intervened Linear output	Confirmed
Score file	<code>sensitivity_robocasa365_atomic_protocolfix_d4_g64_b4_6_dit.json</code> , SHA-256 prefix <code>29e1a5c7a991e2ab</code> , 116 scored linear layers	Present
Allocation entry	Greedy byte-constrained allocation and fixed-size-flip candidate generation; re-running it reproduces the parent plan’s 116-layer vector, byte total, and all nine candidates exactly	Reproduced
Final mask	<code>gr00t_quant_plan_robocasa365_cscka_16tol_adjudicated.final_plan.json</code> , 100 W4/16 FP16, matching candidate 2 (historical label <code>flip8</code>) from the complete-mask adjudication; that ranking stage was not rerun	Present; binds 54.0%

FCP separately uses original-reference physical actions on its stratified buffer. The Gaussian CS bandwidth is fixed from the representation reference; token/step samples are dependent.

A four-task development sweep selects $\lambda_g : \lambda_d = 16 : 1$ by closed-loop macro success, yielding the 100-W4/16-FP16 initial mask. The source statistic definitions remain those cited in Section 3.1.

The full-context stage runs once from the frozen M_0 ; it does not restart from several initial masks and retain the best-performing one. The artifact registry records the exact initializer and calibration-buffer hash. The four ratio-selection tasks remain in the 50-task aggregate. A post-freeze calculation over the other 46 tasks gives 51.8% for ours, 27.6% for QuantVLA W4A8, and 52.8% for FP16, showing that the four development tasks do not create the main pattern.

A.2 EXACT METRIC SPECIFICATION AND EDGE CASES

Sequence construction and action semantics. Each metric unit is a task–seed sequence u with R_u unique replans. We sort replans by their recorded integer index; each contributes the complete inverse-normalized (16, 12) chunk executed before the next replan, so $N_u = 16R_u$. We discard later unexecuted forecasts, do not overlap adjacent chunks, and do not deduplicate predictions. A sequence contains only complete chunks collected before success, termination, truncation, or the official horizon. Duplicate replans, any shape other than (16, 12), or a non-finite value fail the metric gate rather than being padded or imputed. Coordinates 1–3 encode end-effector translation, 4–6 a rotation vector, 7 gripper close, and 8–12 the deployed base/control commands. Equation 6 normalizes each coordinate separately.

For coordinate k , we estimate a robust scale once from the complete FP16 selection buffer and never recompute it per task, sequence, or candidate:

$$\tilde{s}_k = \max(1.4826 \text{ MAD}_k, 0.1 \text{ RMS}_k), \quad s_k = \max(\tilde{s}_k, 0.05 \text{ median}_j \tilde{s}_j, 10^{-4}). \quad (6)$$

With $z_{u,t} = (a_{u,t}^Q - a_{u,t}^T)/s$ and $\rho(z) = 2(\sqrt{1+z^2} - 1)$, the local and accumulated terms are

$$d_{\text{local}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho(z_{u,t,k}), \quad d_{\text{prefix}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho\left(\frac{\sum_{j=1}^t z_{u,j,k}}{\sqrt{t}}\right). \quad (7)$$

Pose map. For $\xi = [v; \omega]$, let $\Omega = [\omega]_{\times}$, $\theta = \|\omega\|_2$, and

$$\begin{aligned} \text{Exp}(\xi) &= \begin{bmatrix} I + A\Omega + B\Omega^2 & (I + B\Omega + C\Omega^2)v \\ 0 & 1 \end{bmatrix}, \\ A &= \frac{\sin \theta}{\theta}, \quad B = \frac{1 - \cos \theta}{\theta^2}, \quad C = \frac{\theta - \sin \theta}{\theta^3}. \end{aligned} \quad (8)$$

The implementation uses Taylor limits for $\theta < 10^{-5}$. Right multiplication, $T_t = T_{t-1} \text{Exp}(\xi_t)$, interprets each increment in the current end-effector frame. The logarithm uses the principal rotation angle and inverse left Jacobian $V^{-1} = I - \Omega/2 + c\Omega^2$, where $c = 1/12$ near zero and $c = \theta^{-2} - (1 + \cos \theta)/(2\theta \sin \theta)$ otherwise. The three control-specific components are

$$e_t^{\text{pose}} = \frac{\text{Diag}(s_{1:6})^{-1} \text{Log}((T_t^T)^{-1} T_t^Q)}{\sqrt{t}}, \quad d_{\text{pose}}^{(u)} = \bar{\rho}_t(e_t^{\text{pose}}), \quad (9)$$

$$d_{\text{stitch}} = \bar{\rho}\left(\frac{(A_{r+1,1}^Q - A_{r,H}^Q) - (A_{r+1,1}^T - A_{r,H}^T)}{s}\right), \quad (10)$$

$$d_{\text{grip}}^{(u)} = \frac{1}{2} \left[\bar{\rho}_t(g_t^Q - g_t^T) + \bar{\rho}_{t>1}((g_t^Q - g_{t-1}^Q) - (g_t^T - g_{t-1}^T)) \right], \quad (11)$$

where $g_t^P = \sigma(4a_{t,7}^P/s_7)$. Translation and rotation are normalized coordinatewise before averaging.

Functional diagnostic. For physical final chunks indexed by context i , define

$$e_i = \frac{\|A_i^Q - A_i^T\|_F^2}{\max(\|A_i^T\|_F^2, 10^{-8})}, \quad d_{\text{final}} = \frac{\|A^Q - A^T\|_F^2}{\max(\|A^T\|_F^2, 10^{-8})}, \quad (12)$$

$$D_{\text{func}} = d_{\text{final}} + d_{\text{kin}} + d_{\text{grip}} + 2 \text{CVaR}_{0.9,i}(e_i).$$

d_{kin} is one half of the sum of analogous relative errors restricted to translation and rotation coordinates. The logged $d_{\text{solver}} = \text{mean}_i e_i$ is a separate summary. In Equation 12, $\text{CVaR}_{0.9,i}(e_i)$ averages the context-level samples e_i at or above their empirical 90th percentile, rather than applying CVaR to that scalar mean. The registered D_{func} gripper term is the fraction of adjacent solver states for which active gripper changes ($> 10^{-4}$ in both policies) disagree in sign. Because the shared selection path records only final physical chunks and not the internal solver trajectory, this legacy D_{func} gripper term is zero on that path. Gripper deviation still enters D_{PAC} through Equation 11 and the explicit component guard.

Evaluation order. For each candidate, we (1) validate paired records, sort replans, and inverse-normalize final chunks; (2) compute the global 12-coordinate scale once from all FP16 selection chunks; (3) evaluate Equations 7, 9, 10, and 11 for each task-seed sequence; (4) form the five-component mean and worst-10% sequence mean in Equation 3; and (5) compute task-level candidate-minus- M_0 differences, the jackknife standard error in Equation 13, J , and the pose/stitch/gripper upper bounds. We set $d_{\text{stitch}} = 0$ for a one-replan sequence and the gripper-difference term to zero for a one-action sequence. Acceptance requires strict $J_{\text{final}} < 0$, non-positive component upper bounds, and the exact byte constraint; equality fails closed.

For n_g task scalars x_t in cluster g , the implemented leave-one-task-out jackknife standard error is

$$\text{SE}^{\text{jack}}(\bar{x}) = \sqrt{\frac{\sum_{t \in g} (x_t - \bar{x})^2}{n_g(n_g - 1)}}, \quad (13)$$

which is algebraically identical to the usual sample standard error of the task mean.

A.3 MASK DIAGNOSTIC ROBUSTNESS

The registered 50-episode diagnostic compares M_0 with uniform full-W4 under an otherwise identical runtime. M_0 obtains 56% versus 46%, with ten paired wins, five losses, and 35 ties ($p = 0.302$). An earlier 50-episode exploratory wave reverses the direction (50% versus 52%); pooling both waves gives 53% versus 49% ($p = 0.557$). We retain this small-sample result as a robustness check outside the headline evaluation.

A.4 PAIRED STATISTICAL DETAIL

Against QuantVLA W4A8, 1,647 of 2,500 episodes are tied, 721 favor DYPAC-VLA, and 132 favor QuantVLA. The exact two-sided McNemar test gives $p = 6.30 \times 10^{-99}$; Holm adjustment over the registered comparison family gives $p = 1.89 \times 10^{-98}$. Against FP16, 1,889 episodes are tied, 292 favor DYPAC-VLA, and 319 favor FP16, giving $p = 0.2929$.

Against the initializer-stage GDSQ-VLA deployment, 1,841 episodes are tied, 370 favor DYPAC-VLA, and 289 favor GDSQ-VLA. The exact McNemar test gives $p = 0.00181$, and Holm adjustment gives $p = 0.00362$. All three split-specific GDSQ-VLA plans use exactly the same 100-W4/16-FP16 decision vector as the final mask. GDSQ-VLA uses the original DuQuant W4 substrate and split-calibrated static A8, whereas the final stack uses Hessian-aware W4 and DYRange-A8. The contrast therefore measures their combined post-allocation effect. In the paired GR00T task-level comparison, ours is higher/tied/lower on 48/1/1 tasks against QuantVLA; Figure 2 shows the common-scope operating points and Table 1 the full comparison with storage scopes marked.

A.5 COMPRESSION-RATE SWEEP

Table 5: GR00T compression-rate sweep of the DYPAC-VLA rate family (50 tasks; seeds 0–9; 500 episodes per arm). Every arm is derived by the same FCP rule from our own pipeline.

Target r	W4/FP16	Linear size (GiB)	Comp.	All-50 SR (%) $\uparrow$	Atomic	C-Seen	C-Unseen
1.20	30/86	1.709	1.166×	55.4	74.4	46.3	43.1
1.60	63/53	1.359	1.466×	52.8	71.7	43.8	40.6
2.00	80/36	1.146	1.739×	54.6	74.4	41.9	45.0
2.40	91/25	1.007	1.979×	51.4	71.7	42.5	37.5
2.80	98/18	0.907	2.196×	54.0	73.9	44.4	41.3
Ours, all-W4 (max)	116/0	0.779	2.556×	55.4	71.7	48.1	44.4

Every arm is one endpoint of our own rate family, built by the frozen FCP ranking from the same 116-layer inventory; none is an external baseline. All arms use Hessian-g64 W4, deployment-conditioned A8, four flow steps, all 50 tasks, and seeds 0–9 (500 episodes per arm). All 3,000 expected units are unique and complete. Target r applies to candidate tensors; size and compression count the audited Linear-component inventory, including fixed components and packing metadata, not the whole-model checkpoint. The frozen 100-W4/16-FP16 headline policy obtains 54.0% at 2.224×

The sweep traces our own rate family rather than an external baseline: every arm is produced by the frozen FCP rule from the same 116-layer inventory, with one frozen layer ordering and the

same Hessian-g64/dynamic-A8 execution stack. Shrinking the static policy by $2.19\times$ across the two endpoints produces no monotonic success degradation: the $2.556\times$ all-W4 endpoint reaches 55.4%, tying the $1.166\times$ arm and exceeding the frozen $2.224\times$ headline rate of 54.0%. The isolated 51.4% trough at $1.979\times$ is driven most strongly by Composite-Unseen, which falls to 37.5% and recovers to 44.4% at the all-W4 endpoint. Across this candidate domain, one pipeline sustains a broad high-success compression plateau.

Formal-scale maximal-rate endpoints. The all-W4 endpoints are additionally evaluated at formal Table-1 scale (50 seeds per task, 2,500 episodes) under the shared four-flow-step protocol and are displayed in Table 1 as DYPAC-VLA (Ours, all-W4). They are the maximal-rate endpoint of the same rate family, not an independently developed baseline. On GR00T, the all-W4 rate-family endpoint reaches 52.8% task-macro success (1319/2500; 70.9/42.9/42.3 by split) at 836,960,256 static bytes ($2.556\times$), versus 54.0% for the audited anchor at $2.22\times$. On $\pi_{0.5}$, the 180-layer all-W4 profile reaches 26.8% (670/2500; 58.7/14.5/3.3 by split) at 1,242,169,344 static bytes ($3.556\times$), versus 27.7% for the projected anchor and 26.2% for FP16. Both rows are descriptive operating points of our own rate family; they are not registered tests, we do not test them against the headline mask, and neither revises the frozen headline rows. The aggregate is content-addressed at `runs/robocasa365_table1_max_sweep_v1/aggregate.json` (SHA256 93bb7b021ca83b073b314c6b18e97f3d7b4c4e327eeb190375d7d5fedb3070bc) and the plans at `runs/robocasa365_table1_max_sweep_v1/masks/gr00t_max.plan.json` (SHA256 b1fe584404184bf6f5505af5ef40ebaeab759a94c1e0e952dd19b085e387179d) and `runs/robocasa365_table1_max_sweep_v1/masks/pi05_max.plan.json` (SHA256 8b0aabc15e48970250e605ca689e08bd e3c9b73616d2c31fd2fae88d01759dbd).

A.6 RIPA SIGNAL ABLATION

Table 6: RIPA signal ablation at identical GR00T bytes (50 tasks; seeds 0–9; 500 episodes per arm).

Signal	W4/FP16	Primary-46 SR (%) $\uparrow$	Fused Δ (pp)	Holm p	All-50 SR (%) $\uparrow$
CS+CKA-DiT (16:1)	93/23	53.26	–	–	55.20
CKA-DiT only	94/22	51.52	+1.74	0.5796	53.20
CS only	92/24	50.65	+2.61	0.5796	52.40

All arms use exactly 962,068,480 static bytes ($2.2239\times$ compression), deployment-conditioned A8, 50 tasks, and seeds 0–9, totaling 1,500 episodes. Primary-46 excludes the four tasks used to select the 16:1 ratio. Deltas are fused minus the single-signal arm; the two paired McNemar tests form one Holm family.

Relation to the headline mask. This is an independent signal comparison under a common exact-byte solver, not a direct reuse of the headline fused mask. All three arms reuse the same frozen sensitivity file, action weights, numerical guards, and 116-layer inventory. The changed allocation entry point is `select_byte_ablation_masks.py`: starting from all-W4, it maximizes protected risk with a binary integer program subject to exactly 125,108,224 extra FP16 bytes. Tie-breaking minimizes protected-layer count and then uses lexicographic order. In contrast, the historical headline parent uses greedy allocation with a Uniform-W6 budget, fixed-size flips, and complete-mask adjudication; its final mask is the `flip8` winner. The equality solver rebuilds the fused arm as 93-W4/23-FP16, versus 100-W4/16-FP16 in Table 1. Both use 962,068,480 static-component bytes because layer sizes differ. Calibration statistics are reused rather than recollected; candidate generation and the equality constraint differ from the headline procedure.

Within this independent comparison, only the allocation signal changes across the three arms, while the exact static byte budget, Hessian-aware W4, deployment-conditioned A8, and rollout protocol stay fixed. The Primary-46 estimand excludes the four tasks used to choose the 16:1 fusion ratio. Historical table and artifact labels CS+CKA-DiT, CKA-DiT only, and CS only denote the fused, relational-only, and distributional-only variants. The fused variant is higher than relational-only on 16 tasks, tied on 17, and lower on 13; against distributional-only the counts are 21/11/14. Both aggregate differences favor fusion descriptively, with Holm-adjusted $p = 0.5796$; a stable gain remains statistically unresolved.

A.7 INDEPENDENT BUDGET-PROJECTION COMPARISON

This completed study isolates the deletion rule in FCP’s budget projection. The protocol fixes two backbones, two budgets each, and three deletion rules, plus one unmodified initializer per backbone: 14 configurations with 50 tasks and seeds 20000–20009, totaling 7,000 episodes. Masks are fixed before these rollouts and use subsets of frozen parent second-order quantization artifacts, without requantization or selection feedback from the new success results. Historical findings motivated the prospective comparison; this does not make the entire research process preregistered.

Matched budgets and controls. Full-policy-score deletion is compared with final-action-error deletion and largest-layer-first deletion. The former simple control uses saved relative squared error of final physical actions, not raw MSE. Both simple rules include subset-sum feasibility corrections to match the full-score configuration’s exact bytes; they are not unconstrained greedy rules. At each budget, the three rules occupy identical static bytes despite possibly different protected-layer counts.

In the order full-policy score / final-action error / largest first, GR00T achieves 51.6/52.2/52.2% at 900,268,032 bytes (95% initializer ceiling) and 54.4/53.8/53.8% at 855,048,192 bytes (90% ceiling). On $\pi_{0.5}$, the rates are 26.6/25.8/25.4% at 1,634,828,288 bytes (original target ceiling) and 25.6/26.0/25.8% at 1,475,051,520 bytes (90% of that ceiling). Full-score deletion removes 7/15 GR00T protections and 41/71 $\pi_{0.5}$ protections. Unmodified initializers achieve 53.6% and 25.0%, respectively, at their larger, unmatched footprints.

Inference and scope. All eight planned exact two-sided McNemar comparisons have raw $p \geq 0.504$ and Holm-adjusted $p = 1.0$. All eight task-clustered bootstrap descriptive 95% success-difference intervals include zero. These results do not establish a stable success advantage for full-score deletion, equivalence of the rules, or absence of small effects. The study permits only removal of initial protections and omits later candidate-state admission, so it is a projection-rule comparison rather than an ablation of all FCP stages. Newly generated $\pi_{0.5}$ masks use corrected statistics: matching a historical layer count and byte size does not establish mask identity, and these results do not replace Table 1.

The protocol is recorded at `scripts/budget_projection_comparison_protocol.json`; the frozen manifest and completed statistics are at `runs/budget_projection_comparison_v1/manifest.json` and `runs/budget_projection_comparison_v1/aggregate.json`. The aggregate records all 14 completed arms, paired comparison counts, confidence intervals, source hashes, and disabled selection feedback.

A.8 REDUCED CLOSED-LOOP COMPONENT ABLATION

The earlier reduced component study uses all 50 tasks and ten paired seeds per task. It is disjoint from mask construction and was aggregated only after all 1,500 new episodes passed the coverage audit. The matched M_0 reference obtains 73.9%, 46.9%, and 38.1% on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively. Its overall success rate is 53.8%.

Table 7: GR00T post-initialization controls (50 tasks; 2,500 episodes for the fixed-mask pair, 500 per control).

Configuration	W4/FP16	Episodes	SR $\uparrow$	Δ (pp)
GDSQ-VLA (DuQuant/static A8)	100/16	2500	50.8	reference
DYPAC-VLA (Hessian/DYRANGE-A8)	100/16	2500	54.0	+3.2
M_0 + DYRANGE-A8	100/16	500	53.8	reference
Frozen-range A8	100/16	500	0.0	-53.8
Local-MSE selection	81/35	500	53.0	-0.8
No final check	115/1	500	53.0	-0.8
FCP proposals (6; diagnostic)	111–115/5–1	3000	50.6–56.2	-3.2–+2.4

The first pair holds the 100/16 mask fixed (Holm $p = 0.0036$). Local-MSE and no-final-check controls each have Holm $p = 1.0$ versus the 500-episode anchor; their distinct masks occupy 1,060,044,800 and 850,526,208 bytes, so they are not an equal-byte, common-candidate admission comparison. The six FCP proposals have minimum Holm $p = 0.891$; Table 9 lists them.

Table 7 collects these controls.

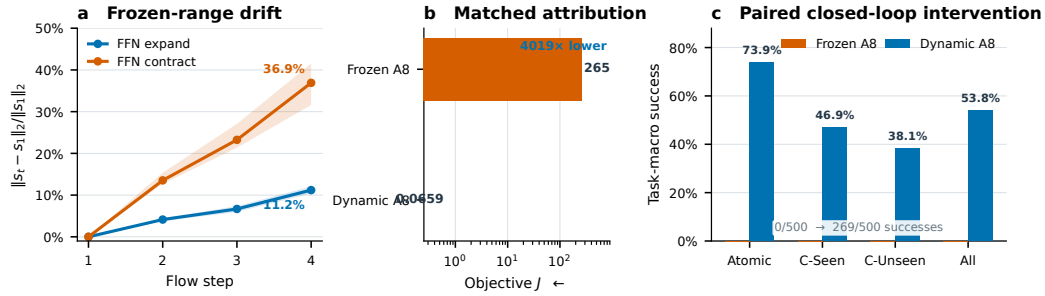


Figure 3: Evidence for per-forward A8 scaling. **(a)** Calibrated per-channel ranges drift across flow steps. **(b)** Matched offline attribution relative to A16 under objective J (log scale). **(c)** In the historical paired control, the registered static table obtains 0/500 successes and dynamic A8 obtains 269/500 under the same M_0 runtime. Table 3 separately reports the fresh-seed confirmation, including a stronger pooled-static baseline.

Frozen-range A8 obtains 0.0% on every split. All 500 episodes complete normally, with no crash or numerical-failure row. Relative to M_0 , it has no paired wins, 269 losses, and 231 ties. Under the registered calibration protocol, range construction alone determines the transition from 0 to 269 successful episodes.

The Local-MSE control selects its mask using local Hessian reconstruction error; this is distinct from the raw physical-action MSE used in the matched-candidate replay. Local-MSE selection obtains 72.8%, 43.8%, and 40.0% across the three splits, for 53.0% overall. Against M_0 , it has 55 paired wins, 59 losses, and 386 ties. The no-final-check proposal obtains 70.6%, 38.8%, and 47.5%, also giving 53.0% overall, with 54 wins, 58 losses, and 388 ties. Their different split behavior—especially the seen/unseen reversal for the no-final-check proposal—shows why the full 50-task result is more informative than results on the two seen groups alone. Equal aggregate rates do not imply equivalent policies.

A.9 DYNAMIC-RANGE EVIDENCE

Figure 3 separates mechanism diagnosis from behavioral evidence. Across the 31 applicable action-head layers, median scale drift reaches 10.8%, 16.8%, and 24.9% at flow steps 2–4; the maximum layer-step value is 49.0%. With the corrected task-level standard error, the registered static table has $J = 264.94$ against the A16 diagnostic. DYRANGE-A8 reduces this discrepancy to $J = 0.066$ and improves over the static table with $J = -0.241$. Under an otherwise identical M_0 runtime, the paired closed-loop intervention gives 0/500 successes for the registered static table and 269/500 for DYRANGE-A8.

A.10 INDEPENDENT STATIC-A8 COVERAGE CONFIRMATION

Table 3 in the main text reports the outcome; this appendix gives the frozen protocol and the complete statistics.

Design and frozen controls. This follow-up evaluates four activation-range rules on all 50 RoboCasa365 tasks with ten new seeds, 23000–23009, per task: 500 episodes per arm and 2,000 in total. Its protocol was frozen after an exploratory 18-task/two-seed pilot but before these new outcomes were collected. The pilot and all historical main/control outcomes are excluded from the new statistics. All four arms share each split’s checkpoint, 100-W4/16-native-precision layout, Hessian W4 codes/scales, identity input transform, four flow steps, execute-16 horizon, fresh environments, and paired action noise. Runtime audits confirm the shared plan, weights, checkpoint, wrapped-layer count, and each arm’s activation table. All 12 split-arm cells complete without error rows.

What changes between the static controls. The registered arm reuses the historical frozen activation table. Two additional static tables are recalibrated from the original FP16 checkpoint separately for each split, using the same 256 calibration observations and paired noise, batch size eight, and

Table 8: Core offline ablations on the frozen 12-task, 36-sequence context buffer; lower objective is better.

Factor	Controlled counterfactual	Audited evidence	Decision
Range	Static A8 vs. FP16-activation control	$J = 264.94$	Reject static A8
Range	Dynamic A8 vs. registered static A8	$J = -0.241$	Use DYRANGE-A8
Proposal audit	FP16-retention value from state flips	2/116 positive retention values	Rebuild protected sets
Policy audit	6 corrected complete policies	0/6 eligible; $J \in [0.95, 2.36]$	Retain M_0

J is the worst normalized one-standard-error upper bound over D_{PAC} , D_{func} , and task clusters, using the corrected jackknife SE of the task mean. Dynamic-vs-A16 has $J = 0.066$. All six corrected candidates have completed teacher-state scoring; none passes the minimax and component guards, so candidate-state scoring is not triggered.

99.9th-percentile scaling per input channel and applicable flow step. The pooled table computes the quantile over all captured rows; the sparse table retains only the first and last flattened row of each forward capture. Thus the dense-sparse comparison changes sampling coverage within the same calibration observations, not the calibration source or number of observations. Both tables remain frozen during evaluation. The label FP16 describes their calibration source; all evaluated policies retain the same mixed W4/native weight layout. The fourth arm uses input-conditioned DYRANGE-A8.

Primary coverage result. Dense static calibration obtains 216/500 successes (43.2%), whereas the matched two-row table and the historical registered table each obtain 0/500. For the pre-frozen primary dense-minus-sparse contrast, there are 216 paired wins, zero losses, and 284 ties: a 43.2-point difference, task-cluster bootstrap 95% CI [35.6, 51.0] points, and exact two-sided McNemar $p = 1.90 \times 10^{-65}$ (Holm-adjusted $p = 2.28 \times 10^{-64}$). The dense-minus-registered comparison has the same counts and statistics. The matched dense-sparse intervention demonstrates that sparse calibration sampling can cause collapse in this setting, while a frozen range table with broader captured-row coverage can recover substantial success.

Dynamic ranges against the stronger static control. Dynamic A8 obtains 269/500 successes (53.8%), with 106 paired wins, 53 losses, and 341 ties against dense static A8. The 10.6-point advantage has 95% CI [6.2, 15.0] points, exact McNemar $p = 3.18 \times 10^{-5}$, and Holm $p = 1.27 \times 10^{-4}$. All 12 pre-frozen comparisons (three contrasts across the overall set and three splits) share one Holm family. Dynamic-minus-dense split differences are 12.8, 10.6, and 8.1 points, with adjusted $p = 0.0155$, 0.0540, and 0.0660, respectively; the latter two are not significant at 0.05. The 95% intervals use 20,000 task-cluster bootstrap replicates with RNG seed 20260907 and are pointwise, not simultaneous family-adjusted intervals.

Implication and provenance. These findings strengthen the evidence for input-conditioned ranges against a functional static baseline, while narrowing the interpretation of the earlier zero-success control to its calibration construction. They establish neither failure of all static schemes nor a separate causal attribution of the remaining dynamic advantage to policy-induced state shift. The new dynamic aggregate happens to equal the historical 269/500 result, but the task-seed records and split outcomes differ and are not pooled. The frozen design and complete summary reside at `runs/static_a8_confirmation_20260908/protocol.json` and `runs/static_a8_confirmation_20260908/summary.json`. The paper renderer verifies their hashes, all raw episode keys and outcomes, runtime lineage, paired counts, and the 12-test Holm calculation; the source summary’s bootstrap intervals were independently reproduced before inclusion.

A.11 FULL-POLICY AUDIT OUTCOME

Figure 4 separates conservative FP16-retention benefit from complete-policy acceptance. Corrected task-level statistics identify two of 116 coordinates with positive conservative benefit for retaining FP16 under both D_{func} and D_{PAC} , but both are already protected: removing their protection increases the proxy loss rather than providing improving flips. The generator rebuilds six complete protected sets containing one to five FP16 layers, not small edits of the 16-layer anchor, and scores all six on

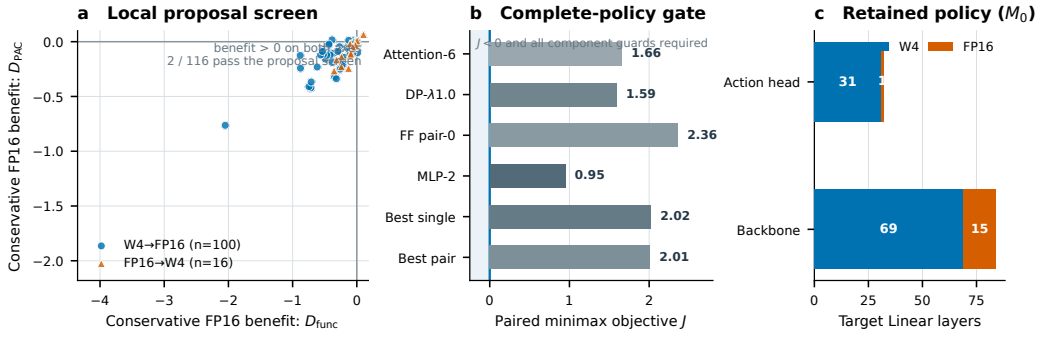


Figure 4: GR00T deviation-based decision record with corrected task-level statistics. **(a)** Two of 116 coordinates have positive conservative benefit for retaining FP16 under both metrics. Both are already protected; these are not two improving removals. **(b)** All six rebuilt complete-policy candidates have $J > 0$ and fail a component guard. **(c)** The 100-W4/16-FP16 initializer is retained. This decision does not establish its success optimality or demonstrate an effect of candidate-state adjudication, which was not triggered.

Table 9: Preregistered post-selection GR00T diagnostic over six FCP proposals (50 tasks; seeds 0–9; 500 episodes per proposal).

Configuration	W4/FP16	SR $\uparrow$	Δ (pp)	W/L	Raw p	Holm p
M_0 (frozen)	100/16	53.8	—	—	—	—
Single-layer	115/1	54.8	+1.0	59/54	0.707	1.000
Two-layer	114/2	50.6	-3.2	46/62	0.149	0.891
Attention block 6	112/4	55.6	+1.8	56/47	0.431	1.000
MLP block 2	113/3	53.0	-0.8	53/57	0.775	1.000
FF pair	114/2	56.2	+2.4	59/47	0.285	1.000
DP-λ1.0	111/5	52.8	-1.0	54/59	0.707	1.000

Each proposal uses 50 tasks and ten seeds per task (500 episodes); Δ and paired wins/losses are relative to the same 500-episode M_0 control. The six tests were fixed before rollout and adjusted as one Holm family.

teacher-state contexts over the three registered splits. Their objectives lie in $[0.948, 2.360]$, and each fails at least one component guard. The protocol retains M_0 without candidate-state scoring; this decision neither exhausts the feasible neighborhood nor establishes superiority in task success, and no prior rollout, byte count, or checkpoint is changed. Local retention scores do not establish that removing other protections simultaneously will yield an acceptable configuration.

The candidate set, task–seed coverage, paired tests, and Holm family were frozen before outcomes were revealed, and the preregistration forbids feeding these results back into selection. The largest observed difference is +2.4 points for the FF-pair proposal (raw $p = 0.285$, Holm $p = 1.000$); the minimum adjusted value across the family is 0.891. These post-selection outcomes leave task-success differences unresolved; they do not validate rejection as preventing a regression. Outcomes are reported without retrospectively selecting the most favorable rollout estimate.

The final plan uses signed-nibble group-64 weights, identity row rotation, and per-forward dynamic A8, with no ATM, OHB, error folding, calibration bypass, or runtime selector. Its SHA-256 is recorded in the artifact registry and checked before table generation.

A.12 FCP PROJECTION, ADJUDICATION, AND FREEZE RULES

Evidence-guided feasibility projection. If $C(M_{\text{init}}) \leq C_{\text{max}}$, FCP sets $M_0 = M_{\text{init}}$. Otherwise, it instantiates each FP16-to-W4 removal as a complete policy and ranks removals by their deviation evidence. For shared context $\mathcal{B}$, a coordinate intervention has the form

$$\Delta_i(M) = D(Q_{M^{(i)}}; \mathcal{B}) - D(Q_M; \mathcal{B}), \quad (14)$$

where $M^{(i)}$ changes target i and D denotes the applicable full-policy deviation criterion. The intervention is measured in the current mixed-precision background rather than as isolated layer

reconstruction, and protections are removed from least to most damaging until the mask is feasible. This projection uses ranked single-layer evidence, not an exact solution of all multi-layer interactions.

Coordinate counterfactuals and the candidate class. Starting from M_0 , FCP flips target precision states one at a time and, when scheduled, repeats the sweep around the preceding winner. Each intervention is instantiated as a mixed-precision network and compared with its complete-mask background, and the resulting conservative benefits feed an exact sparse Pareto 0–1 dynamic program subject to

$$C(M) = C_{\text{fixed}} + \sum_{i \in \mathcal{L}} C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}. \quad (15)$$

The constraint is expressed in bytes rather than layer count because layers differ in tensor and metadata size. The dynamic program only generates candidates: its additive benefits cannot capture interactions among simultaneous layer changes. The finite class $\mathcal{C}(M_0)$ contains the anchor, dynamic-program proposals, and five structured proposal types rebuilt from layer scores, and full-policy evaluation checks these complete configurations.

Task-level minimax adjudication. For metric $q \in \{D_{\text{func}}, D_{\text{PAC}}\}$ and task cluster $g \in \{\text{all, Atomic, C-Seen, C-Unseen}\}$, let $\Delta_{q,t}(M)$ be the paired candidate-minus- M_0 task scalar. We define

$$U_{q,g}(M) = \frac{\text{mean}_{t \in g} \Delta_{q,t}(M) + \text{SE}_{t \in g}^{\text{jack}}[\Delta_{q,t}(M)]}{\text{mean}_{t \in g} q_t(M_{\text{base}})}, \quad J(M) = \max_{q,g} U_{q,g}(M). \quad (16)$$

Here $M_{\text{base}} = M_0$ shares the candidate’s context, and the jackknife uses tasks as uncertainty units. With positive denominators, $J(M) < 0$ requires *strict one-SE improvement in every metric–group pair*, plus non-positive component bounds for pose, stitch, and gripper. The rule preserves the anchor unless deviation evidence favors replacement: extra compression alone cannot justify admission. It is frozen after evaluation and supplies no safety or success guarantee.

Candidate-state check and freeze. The protocol reserves candidate-induced observations over four to eight replans and defines $J_{\text{final}} = \max(J_{\text{teacher}}, J_{\text{candidate}})$. Eligible teacher-state candidates are first filtered by a paired one-standard-error comparison against the best teacher-state objective; up to three then enter candidate-state adjudication. Let $\mathcal{E}(M_0)$ denote the remaining byte-feasible candidates with negative J_{final} and the required deviation guards. The frozen choice is

$$\widehat{M} = \begin{cases} \text{lexmin}_{M \in \mathcal{E}(M_0)}(C(M), N_{16}(M), \text{names}(M)), & \mathcal{E}(M_0) \neq \emptyset, \\ M_0, & \mathcal{E}(M_0) = \emptyset, \end{cases} \quad (17)$$

where N_{16} counts protected layers and names break ties deterministically. Storage is compared only after deviation eligibility. On GR00T all proposals fail the teacher-state screen, so FCP retains M_0 without candidate-state evaluation or any success-optimality claim.

A.13 REAL-HARDWARE MEASUREMENT DETAIL

Relative to native FP16, M_0 reduces median CUDA peak allocation from 5.386 to 4.267 GiB ($0.792\times$) and median NVML device use from 7,644 to 6,720 MiB. Median latency nevertheless rises from 107.5 to 232.2 ms ($2.16\times$), and gross board energy rises from 18.65 ± 4.23 to 39.81 ± 0.46 J/request ($2.14\times$). The six rejected candidates span 238.2–245.8 ms, 4.165–4.218 GiB, and 40.45–41.58 J/request. The packed representation therefore provides a measured memory benefit, but neither static bytes nor W4-layer count predicts speed or energy under this generic nibble-dequantize-then-FP16-GEMM backend.

Real-integer backend revalidation. We next replace the generic execution path with packed W4 residency, dynamic signed A8, and $\text{INT8} \times \text{INT8} \rightarrow \text{INT32}$ GEMM. The runtime-contract audit verifies 4-bit weights, 8-bit activations, integer GEMM, the absence of an FP-weight-sized buffer, 409,468,928 packed-weight bytes, and consistent metadata across all three task splits. Kernel compatibility changes activation scaling from the frozen policy’s per-input-channel rule to per-row group-64. We therefore treat this result as a separately named post-hoc deployment point that cannot reopen FCP.

Table 10: GR00T hardware measurements on an idle NVIDIA A40 at batch size one (three 60-request trials per configuration; lower is better).

Configuration	W4	p50 (ms)	p95 (ms)	Peak alloc. (GiB)	Gross J/req.	Idle-adj. J/req.
Native FP16	0	107.5	143.2	5.386	18.65±4.23	10.65
Initializer stack	100	269.6	297.2	4.319	40.76±1.15	21.52
M_0 (ours)	100	232.2	258.3	4.267	39.81±0.46	22.77
Single-best	115	245.2	271.0	4.165	41.58±0.24	23.68
Two-best	114	245.8	268.4	4.171	41.40±0.41	23.42
Attention-6	112	245.5	258.0	4.165	41.32±0.27	23.60
MLP-2	113	240.6	261.2	4.200	40.45±0.23	22.76
FF-pair-0	114	244.4	267.5	4.172	41.48±0.57	23.46
DP- λ 1.0	111	238.2	293.5	4.218	40.94±0.84	23.17
M_0 paired reference	100	229.7	239.4	4.267	39.25±0.24	22.65
M_0 real W4A8 integer	100	187.5	199.3	4.267	34.64±0.22	20.97

Initializer stack is the 100-W4/16-FP16 DuQuant/static-A8 control (historical run ID quantvla.w4a8), not Table 1’s 116-W4 QuantVLA. Each cell aggregates three trials of 60 measured requests after ten warmups (180 requests/configuration), with four flow steps and CUDA synchronization. p50 is the median of trial medians; p95 is pooled. Peak allocation is the median trial CUDA peak. Board energy is trapezoidal integration of 10-Hz NVML power samples; idle-adjusted values subtract a five-second pre-trial baseline. The paired reference and real-integer row must be compared with each other, not across measurement waves. Measurements exclude model loading, simulator, transport, host energy, and robot energy.

The complete paired closed-loop rerun contains no crash, missing key, duplicate, or terminal failure. The integer backend records 251/500 successes (50.2%) versus 269/500 (53.8%) for the frozen reference: -3.6 points, with 36/54/410 wins/losses/ties, exact two-sided McNemar $p = 0.07255$, and a 10,000-draw task-cluster bootstrap interval of $[-8.8, +1.6]$ points (seed 20260903). The original fake-quant operating point remains the formal policy result, while this row quantifies the separately named integer deployment backend.

In a separate paired hardware wave, the real-integer backend reduces p50 latency from 229.7 to 187.5 ms (18.4%) and pooled p95 from 239.4 to 199.3 ms. Gross board energy falls from 39.25 ± 0.24 to 34.64 ± 0.22 J/request (11.8%), while idle-adjusted energy falls from 22.65 ± 0.35 to 20.97 ± 0.19 J/request (7.4%). Median peak CUDA allocation remains 4.267 GiB. Each row uses three 60-request trials after ten warmups on the same otherwise idle A40. These model-only measurements exclude simulator, transport, host, and robot energy.

A.14 $\pi_{0.5}$ FCP DIAGNOSTIC AND FORMAL RESULT

The transferred $\pi_{0.5}$ initializer contains 80 W4/100 FP16 layers and occupies 1,845,100,544 bytes, exceeding the 1,639,513,497-byte ceiling. FCP evaluates all 180 one-layer precision-state interventions in complete-policy context, then converts the 41 least damaging FP16 protections to W4. The resulting 121-W4/59-FP16 anchor occupies 1,634,828,288 bytes. Subsequent exact-byte proposals do not clear the minimax and component guards, so the projected anchor becomes the final policy.

Table 11: $\pi_{0.5}$ FCP budget-projection diagnostic (50 tasks; seeds 0–9; 500 episodes per mask).

Configuration	W4/FP16	Size (GiB)	SR $\uparrow$	Δ (pp)	Paired p	Status
M_{init} (transferred)	80/100	1.718	26.0	-2.2	0.235	Over budget
Projected M_0	121/59	1.523	28.2	reference	–	Deployed
Single-best proposal	179/1	1.202	24.8	-3.4	0.086	Rejected
Two-best proposal	178/2	1.247	25.0	-3.2	0.086	Rejected

Each configuration uses the same 50 tasks and seeds 0–9 (500 paired episodes). Deltas are relative to projected M_0 . The projection comparison is the preregistered singleton primary test ($p = 0.235$); the two proposal values are Holm-adjusted. The frozen choice is projected M_0 .

A separately preregistered, same-protocol diagnostic evaluates all four masks on the same 50 tasks and seeds 0–9. The projected anchor records 141/500 successes versus 130/500 for the transferred initializer, with 41/30/429 paired wins/losses/ties and exact two-sided McNemar $p = 0.235$. Thus the

projection meets the exact byte constraint while the observed success difference favors the projected mask; the test establishes neither equivalence nor non-inferiority. The single- and two-best proposals record 124/500 and 125/500 successes; both are lower than the anchor after the frozen selection.

The frozen policy uses four flow steps on the complete RoboCasa365 target split, matching the completed reference rows, with 50 seeds on each of 50 tasks. It completes all 2,500 episodes with no missing, duplicate, or failed row. The policy records 536/900 Atomic-Seen, 123/800 Composite-Seen, and 34/800 Composite-Unseen successes, for 693/2,500 overall; the corresponding task-macro rates are 59.6%, 15.4%, 4.3%, and 27.7%. This establishes the paper’s primary success–storage operating point at $2.702\times$ compression.

A.15 EXACT-BUDGET OFFLINE PROXY PREDICTIVE VALIDITY

As an auxiliary diagnostic, we test whether frozen offline scores rank the closed-loop consequences of precision allocation. The library contains 60 masks with identical 100-W4/16-FP16 allocation and exact static bytes, stratified into ten masks at each of six swap distances from M_0 . We freeze offline scores before running one rollout per target task and mask. Across all masks, Spearman correlation with signed success degradation is -0.116 for D_{PAC} and -0.028 for MSE. Mask distance confounds this pooled comparison: larger swaps have greater offline distortion but, in this library, slightly higher mean success. Within each fixed-distance stratum, the registered D_{PAC} correlations are positive but imprecise ($\rho = 0.055\text{--}0.352$, ten masks per stratum), and D_{PAC} does not consistently outperform MSE or D_{func} . These results position D_{PAC} as a predeclared complete-policy deviation guard rather than a success-ranking objective.

A.16 LOCAL CROSS-ARCHITECTURE ADAPTATIONS

ActQuant. We pin the official ActQuant implementation at commit `b64791125070652fe6b554e244fe809c79ef5246` and retain its allocation and quantization semantics (Akbari et al., 2026). Because local adapters provide the GR00T, $\pi_{0.5}$, and RoboCasa365 integration, we label the result `source_protocol_equivalent=false`. Calibration uses an FP16-teacher proxy and excludes formal seeds 0–49; no formal outcome selects or tunes the packs. Both ActQuant configurations complete all 50 tasks and seeds 0–49 under the four-flow-step protocol. GR00T records 649/900, 341/800, and 332/800 successes on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively, for 1,322/2,500 (52.9%) overall. The corresponding $\pi_{0.5}$ counts are 509/900, 98/800, and 22/800, for 629/2,500 (25.2%). Against DYPAC-VLA, ActQuant has 340/368 paired wins/losses on GR00T (Holm $p = 0.3102$) and 135/199 on $\pi_{0.5}$ (Holm $p = 0.0011$). Its measured static packs occupy 3.655 GiB and 2.904 GiB, respectively; these sizes do not imply runtime-memory or latency gains.

QVLA-code. We cloned the official QVLA GitHub repository at commit `26cc4821a3be4c003d09d3c7997b38db2a347982` and followed its released Hessian proxy, channel-gate set $\{0, 2, 4, 8, 16\}$, greedy allocation rule, and row-wise fake-quantization semantics (Xu et al., 2026b). The release provides QVLA quantization and evaluation paths for OpenVLA-family models on LIBERO, but not GR00T N1.5 or $\pi_{0.5}$ tensor mappings, a deployed mixed-row exporter for these architectures, or a RoboCasa365 evaluation path. We therefore add local cross-architecture adapters and label the run `source_protocol_equivalent=false`. Under the shared four-flow-step RoboCasa365 protocol, the content-bound run produces 0 successes in 1,728 completed GR00T episodes and 0 successes in 1,517 completed $\pi_{0.5}$ episodes, despite passing local source-parity, pack-loading, and runtime-metadata checks. We stop the run and omit QVLA-code from Table 1. Because the public repository lacks the target-architecture and target-benchmark paths, the outcome is recorded as a failed cross-architecture reproduction; the native OpenVLA/LIBERO result is unaffected.

DA-PTQ. DA-PTQ reports CogACT W4A8 mixed-precision experiments calibrated with 512 BridgeData V2 trajectories, together with 42.5% memory reduction and 54.8% speedup in its WidowX setup. These measurements use a different model, benchmark, calibration set, hardware path, and accounting scope from our eager fake-quant RoboCasa365 experiments. Table 12 therefore presents them only as source-reported context, without numerically ranking them against our local results. Our frozen clean-room adaptation follows the paper-level allocation and transformation semantics

while adding GR00T, $\pi_{0.5}$, and RoboCasa365 mappings absent from the public release. Both local configurations complete all 50 tasks and seeds 0–49 under the four-flow-step protocol. GR00T records 73/900, 0/800, and 3/800 successes on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively, for 76/2,500 (3.0%) overall. The corresponding $\pi_{0.5}$ counts are 527/900, 98/800, and 30/800, for 655/2,500 (26.2%). Against DYPAC-VLA, DA-PTQ has 8/1,282 paired wins/losses on GR00T and 182/220 on $\pi_{0.5}$; the latter comparison has Holm $p = 0.0648$. We report these results as a complete local cross-architecture adaptation with `source_protocol_equivalent=false`, not as a reproduction of the native CogACT/SimplerEnv results.

Table 12: Source-reported DA-PTQ task-average success on CogACT/SimplerEnv (not locally reproduced).

SimplerEnv setting	CogACT FP	VLA-Cache	QuantVLA	DA-PTQ
WidowX visual matching	51.3	46.8	43.5	48.9
Google Robot visual matching	74.8	74.4	65.6	68.5
Google Robot variant aggregation	61.3	62.3	49.4	51.7

Values are transcribed from Tables 2–3 of the cited paper. Different models, task distributions, calibration, and evaluation protocols preclude a numerical ranking against Table 1.

A.17 STORAGE ACCOUNTING AND NATIVE REFERENCE SCOPES

Table 1’s common reference inventories contain 2,139,537,408 bytes for GR00T and 4,416,602,112 bytes for $\pi_{0.5}$. These are declared Linear-component counts, not whole-model checkpoint sizes; vision modules, activations and runtime caches are outside the scope. GR00T’s fixed 327,352,320 bytes cover non-candidate components within that inventory, and its final plan occupies 962,068,480 bytes ($2.223\times$), or $0.9982\times$ the 963,772,416-byte QuantVLA cell, within the preregistered $1.10\times$ ceiling; the $\pi_{0.5}$ plan occupies 1,634,828,288 bytes ($2.7016\times$). Common-scope compression divides the corresponding exact FP16 bytes by each row’s component bytes, before display rounding. These are static packed estimates, so the ratios alone imply no runtime benefit; Table 10 measures model-only latency, peak memory, and board energy directly instead.

Component scope versus whole-checkpoint payload. The audited inventory is a strict subset of each checkpoint’s stored tensors. Each of GR00T’s three split checkpoints holds 7,585,939,328 bytes (7.065 GiB) across 899 tensors, so the 2,139,537,408-byte inventory covers 28.2% of that payload and 5,446,401,920 bytes (5.072 GiB) lie outside the audited scope and are unchanged by the plan. Every one of the plan’s 116 layers resolves to a checkpoint tensor, and re-deriving their 16-bit payload gives 1,811,939,328 bytes, within 245,760 bytes of the 1,812,185,088-byte candidate anchor. Substituting the 962,068,480-byte quantized components into the same payload gives 6,408,470,400 bytes, so the complete checkpoint corresponds to $1.18\times$ storage reduction while the audited components give $2.22\times$. For unchanged out-of-scope bytes $U > 0$, $(C_{FP} + U)/(C_Q + U) < C_{FP}/C_Q$ whenever $C_{FP} > C_Q$, so component-scope ratios are upper bounds rather than whole-model claims. We do not report an equivalent $\pi_{0.5}$ payload because its frozen inventory and the available checkpoint trees are not reconciled to the same standard. The measurement is recorded at `scripts/tools/audit_checkpoint_scope.py` and `docs/gdsq_vla_iclr2027/evidence/checkpoint_scope_audit.json`.

ActQuant uses a broader whole-model pack scope: its GR00T reference is 5,448,327,040 bytes (5.074 GiB) per checkpoint, versus a mean pack of 3.655 GiB across three split checkpoints, giving native-scope compression $1.388\times$. On $\pi_{0.5}$ its reference is 6,714,075,840 bytes (6.253 GiB), with a 3,117,743,968-byte pack (2.904 GiB), giving $2.154\times$. Those native ratios cannot be combined with the common FP16 row. We mark these sizes P and suppress the common-scope compression cells instead of dividing incomparable inventories.

The $\pi_{0.5}$ Ω -QVLA calculator covers 252 PaliGemma/expert Linear weights and biases, with reference 4,586,471,424 bytes (4.271 GiB). Its tightly packed deployment estimate is 1,402,970,112 bytes (1.307 GiB), including weights, scales, block rotations, and permutation indices, giving native compression $3.269\times$. The evaluated pack contains dequantized caches and is larger than this deployment estimate. This row is marked L and its common-scope compression suppressed. GR00T’s

Table 13: Controlled temporal probes of the production D-PAC implementation (deterministic metric tests; no robot rollouts).

Perturbation	Raw MSE	Local	Prefix	Stitch	Grip
Persistent bias	2.08e-06	0.00467	0.109	0	0
Alternation	2.08e-06	0.00467	0.0002	0.018	0
Boundary impulses	1.95e-07	0.000438	8.48e-06	0.018	0
Interior impulses	1.95e-07	0.000438	1.42e-05	0	0
Gripper delay	8.33e-06	0.012	0.038	0.0641	0.0362
Base-coordinate control	8.33e-06	0.012	0.038	0.0641	0

Ω -QVLA calculator uses the common 2,139,537,408-byte reference and remains in the common-scope plot. Raw native measurements remain unchanged in their source artifacts.

The provenance is `scripts/tools/quantvla_table1_bytes.py`, `runs/qvla_actquant_table1/formal/aggregate.json`, and `runs/gdsq_extension_preregistered_v1/omega_qvla_pi05_robotcasa365_v1/aggregate/pi05_paper_memory.json`. Success comparisons retain all evaluated methods; storage comparisons distinguish these scopes.

A.18 CONTROLLED TEMPORAL PROBES AND MATCHED-CANDIDATE REPLAY

The CPU-only validation calls the production five-component implementation in `quantvla_metric_protocol.py`. Each sequence contains four 16-action chunks with 12 coordinates. A separate sinusoidal reference buffer supplies one fixed coordinate scale. The paired tests fix raw MSE and local error while comparing persistent versus alternating translation error, equal impulses at chunk boundaries versus inside chunks, and a four-step delayed gripper transition versus the same error on a base coordinate. The expected component directions and equality assertions are specified in the test before execution. All three paired checks pass. This establishes responses to these controlled inputs, not closed-loop gains or universal discrimination.

A separate retrospective replay holds the six existing GR00T candidate masks, byte ceiling, teacher-state observations, and one-SE strict-improvement rule fixed. It compares raw physical-action MSE alone, D-PAC alone, and D-PAC with component conditions. All three rules reject all six candidates and select the same anchor. MSE task means are reconstructed from the three original split score files: the merged file retains only one split’s MSE summary and is not used for that comparison. Every replay rule uses the same 12 tasks. The D-PAC bounds and component conditions are taken from the corrected frozen report. Consequently this candidate set reveals no incremental decision value for the more elaborate rule; no new candidate-state evaluation or independent success benefit is inferred. The Local-MSE/no-final-check closed-loop controls in Table 7 are a different experiment, not newly selected replay outcomes.

`scripts/tools/validate_dypac_method_evidence.py` reproduces both checks without loading a VLA or simulator. The output `docs/gdsq_vla_iclr2027/evidence/method_validation.json` records the inputs’ hashes, per-component values, per-candidate bounds, and selected masks.

A.19 AVAILABLE OFFLINE CONFIGURATION COSTS

The 175.87 seconds sum the per-mask `elapsed_s` records over three split files for 1,008 policy-observation evaluations. This is measured scoring time, excluding model loading, coordinate construction, calibration collection, and weight packing. It is not total FCP wall time. The 144 selected observations likewise do not count every environment step needed to reach the sampled windows. Complete stage-wide wall time, total collection steps, and end-to-end configuration cost cannot be recovered from these records and remain unreported. On $\pi_{0.5}$, the projection inventory has 180 interventions and removes 41 protections; no comparable complete timing record is claimed.

Table 14: Offline work recoverable from the frozen GR00T records.

Stage	Forward-work record	Environment work	Time (s)
RIPA bank	116 targets; 16 representation inputs; 8×2 action contexts	Synthetic RoboCasa365 format; see A.1	NR
RIPA adjudication	Nine masks, 16 paired inputs per ratio	Cached inputs	NR
W4 construction	256 calibration observations per W4 layer	Reused calibration	NR
FCP coordinates	116 one-layer interventions on shared context	12 tasks, 36 sequences, 144 selected observations	NR
FCP teacher screen	Seven masks $\times$ 144 observations; four flow steps	Cached observations; no new rollouts	175.87
Candidate-state branch	Zero admitted candidates on GR00T	Zero candidate-state rollouts	Not run

NR denotes a stage-wide measurement that was not recorded. Counts distinguish calibration inputs from policy evaluations.

A.20 LIBERO PROTOCOL DETAIL

Table 15: LIBERO per-suite results on $\pi_{0.5}$ (10 tasks per suite; 100 episodes per suite).

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$	Size (GiB) $\downarrow$
$\pi_{0.5}$						
FP16	95.0	100.0	98.0	94.0	96.8	4.113
QUANTVLA W4A8	97.0	97.0	98.0	93.0	96.3	1.388
Uniform W6	99.0	97.0	99.0	92.0	96.8	1.902
Ω -QVLA W4A4	100.0	99.0	97.0	96.0	98.0	1.307 ^L
QVLA (4.0 BPW) [†]	96.4	98.0	97.2	91.8	95.8	2.7 GB
ActQuant (4.0 BPW) [†]	96.8	98.4	99.4	91.8	96.6	2.7 GB
DyPAC-VLA (Ours)	99.0	100.0	96.0	94.0	97.3	1.788

Table 15: LIBERO per-suite results on GR00T N1.5 (10 tasks per suite; 100 episodes per suite).

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$	Size (GiB) $\downarrow$
GR00T N1.5						
FP16	88.0	87.0	96.0	76.0	86.8	1.993
QUANTVLA W4A8	89.0	85.0	79.0	72.0	81.3	0.898
Uniform W6	85.0	83.0	92.0	22.0	70.5	1.109
Ω -QVLA W4A4	87.0	86.0	89.0	76.0	84.5	0.599
DyPAC-VLA (Ours)	97.0	95.0	98.0	85.0	93.8	0.948

The complete local matrix contains 4,000 episodes across 40/40 exact-valid cells. Each local suite uses ten tasks and ten held-out initial states. Size includes packed weights and metadata and is averaged over suite-specific masks; sizes count the audited Linear-component inventory, not whole-model checkpoints. ^LThe $\pi_{0.5}$ Ω -QVLA size is a distinct 252-Linear inventory and is not comparable with the common component scope (Appendix A.17). [†]The $\pi_{0.5}$ QVLA and ActQuant rows are source-reported 4.0-BPW results (Akbari et al., 2026) using the source memory definition, reported in GB.

Table 15 reports the complete suite matrix. Each local cell contains 100 episodes over ten tasks and held-out states 10–19, and all 40 cells pass the joint gate. The corrected GR00T rerun uses five actions per replan and paired suite-keyed noise; we do not interpret the correction as an algorithmic gain. Local size is the average packed weight and metadata footprint across suite-specific masks. Thus LIBERO transfer concerns the framework and execution rules; the masks are separately calibrated for each suite and then remain fixed during deployment, rather than reusing the RoboCasa365 masks unchanged. The two source-reported 4.0-BPW $\pi_{0.5}$ anchors (Akbari et al., 2026; Xu et al., 2026b) are excluded from the 4,000-episode gate, and their 2.7-GB memory measure is not directly compared with our audited static size.